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A Mini Project Report on

SMART DUSTBIN USING ARDUINO UNO

Submitted in partial fulfillment of the requirements for the award of the degree of

Bachelor of Engineering

in

Electronics and Communication Engineering

by

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CERTIFICATE

This is to certify that the project work entitled **"SMART DUSTBIN USING ARDUINO UNO"** is carried out by **NAGASHREE A M (1KI21EC060), SUCHITRA REDDY K (1KI21EC091), UMASHANKAR G V (1KI21EC106), YASHAVANTH H N(1KI22EC417)** , the bonafide students of Kalpataru Institute of Technology, Tiptur in partial fulfillment for the award of "Bachelor of Engineering" in department of "Electronics and Communication Engineering" of the Visvesvaraya Technological University, Belagavi, during the year 2021-2024. It is certified that all the corrections/suggestions indicated for internal assessment have been incorporated in the report deposited in the departmental library. The project report has been approved as it satisfies the academic requirements in respect of project work prescribed for the said degree.

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ABSTRACT

As people are getting smarter, so are the things. While the thought comes up for Smart cities. There is a requirement for Smart waste management. It is a common sight to witness garbage spilled out in and around the dustbins. The area around an improperly maintained dust bins can house disease spreading insects like mosquitoes, flies, bees and driver ants. The environment around a dustbin is also conducive for increasing the pollution level in air. Air pollution due to a dustbin can produce bacteria and virus which can produce life threatening diseases in human beings. The idea of Smart Dustbin is for the Smart buildings, Colleges, Hospitals and Bus stands. The Smart Dustbin thus thought is an improvement of normal dustbin by elevating it to be smart using sensors and logics. For Smart dustbin operation we are using ultrasonic sensor for detecting distance and object and another sensor servomotor is used for opening and closing the dustbin top.[1]

Keywords: Ultrasonic Sensor, Arduino Uno, Servo Motor

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Chapter 1

INTRODUCTION

The rate increasing population in our country has increasing rapidly and also, we have increase in garbage which have increased environmental issue. Dustbin is a container which collects garbage's or stores items which recyclable or non-recyclable, decompose and non- decompose. They are usually used in homes, office etc, but in case they are full no one is there to clean it and the garbage are spilled out. The surrounding of a dustbin is also conducive for increasing the pollution level . Air pollution due to a dustbin can produce bacteria and virus which can produce life harmful diseases for human. Therefore, we have designed a smart dustbin using ARDUINO UNO, ultrasonic sensor which will sense the item to be thrown in the dustbin and open the lid with the help of the motor. It is an IOT based project that will bring a new and smart way of cleanliness. It is a decent gadget to make your home clean, due to practically all offspring of home consistently make it grimy and spread litter to a great extent by electronics, rappers and various other things. Since the smart dustbin is additionally intriguing and children make fun with it so it will help to maintain cleanliness in home. It will be applied for various type of waste. Dustbin will open its lid when someone/object is near at some range then it will wait for given time period than it will close automatically. Here lid will close when you don't want to use and it will only open when it required.as well as inside the city with keeping all security aspects in mind. These systems will also help us in waste management. The authorities would also have a track record of each and every user which would help them to monitor the cleanliness in the city user which would help them.[2]

1.1 Objective

- The Main Objective of the project is to design a smart dustbin which will help in keeping environment clean and also eco friendly.
- Nowdays Technologies are getting smarter day-by-day so as to clean the environment we are designing a smart dustbin by using Arduino .
- The smart dustbin management system is built on microcontroller based system having ultrasonic sensors on the dustbin.
- To detect when a user approaches the dustbin automatically open the lid and automatically close the lid by using a sensor.
- This hands-free operation not only promotes hygiene but also minimizes direct contact, thereby reducing the spread of germs.
- Provide a convenient and modern way to use a dustbin, making it easier for users to dispose of waste.

Chapter 2

PROPOSED SYSTEM

Explain proposed system with block diagram.

2.1 Methodology

Smart Dustbin is an ARDUINO based project. Here we are using Arduino for code execution, for sensing we used ultrasonic sensor which will open lid and wait for few moment. It will bring drastic changes in tern of cleanliness with the help of technology. Everything is getting with smart technology for the betterment of human being. So this help in maintaining the environment clean with the help of technology. It is a sensor based dustbin so it would be easy to access/use for any age group. Our aim is also to make it cost effective so that many numbers of people can get the benefit from this.

The smart dustbin system leverages the power of Arduino programming to deliver an efficient waste management solution. The ultrasonic sensor plays a pivotal role in detecting the presence of a user, triggering the servo motor to open the lid without the need for physical contact. This automation not only enhances hygiene but also provides an accessible solution for people of all ages, including children and the elderly. Moreover, the system can be further enhanced by integrating additional sensors or IoT modules to monitor fill levels and send alerts for timely waste collection. By focusing on cost-effectiveness, the smart dustbin can be manufactured and distributed widely, encouraging widespread adoption and contributing to cleaner and healthier communities. This project exemplifies how modern technology can be harnessed to create practical solutions that improve daily life and promote sustainability.

2.2 Block Diagram

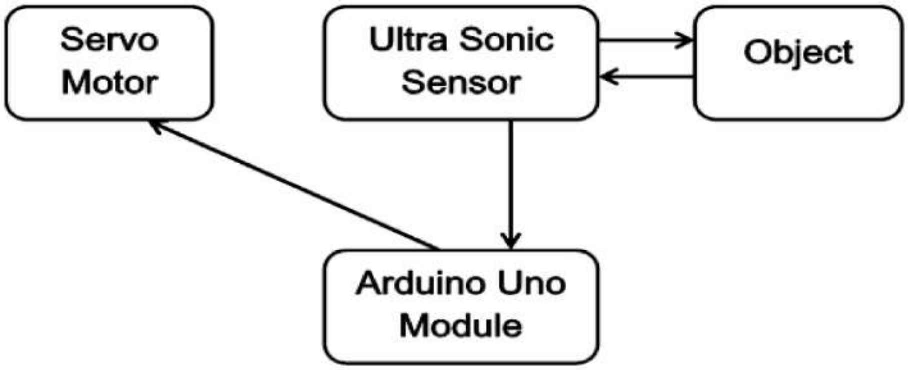

Figure 2.1: Block Diagram

The block diagram for a smart dustbin using Arduino typically consists of these main blocks:

- Arduino Uno: This block represents the microcontroller brain of the smart dustbin. It reads sensor data, processes it according to the programmed code, and sends control signals to actuators.
- Ultrasonic Sensor : This block represents a sensor, typically an ultrasonic sensor (HC-SR04) in this case. It emits high-frequency sound waves and measures the time it takes for the echo to return. Based on this time, it calculates the distance to an object approaching the dustbin.
- Servo Motor: This block represents a motor that controls the opening and closing of the dustbin lid. The Arduino sends control signals to the servo motor, specifying the desired position for the lid (open or closed).

Chapter 3

SYSTEM DESIGN

Smart Dustbin using Arduino Uno system consists of different hardware components. These hardware components are discussed below:

3.1 Arduino Uno Board

Figure 3.1: Arduino Board

The Arduino UNO is a microcontroller board based on the ATmega328. It has 14 digital input/output pins (of which 6 can be used as PWM (pulse width modulation) outputs), 6 analog inputs, a 16 MHz ceramic resonator, a USB (universal serial box) connection, a power jack, an ICSP (in-circuit serial programming) header, and a reset button. It contains everything needed to support the microcontroller; simply connect it to a computer with a USB cable or power it with an AC-to-DC adapter or battery to get started.

3.2 Servomotor

Figure 3.2: Servo Motor

A servo motor is one of the widely used variable speed drives in industrial production and process automation and building technology worldwide. Although servo motors are not a specific class of motor, they are intended and designed to use in motion control applications which require high accuracy positioning, quick reversing and exceptional performance. These are widely used in robotics, radar systems, automated manufacturing systems, machine tools, computers, CNC machines, tracking systems, etc

3.3 Ultra Sonic Sensor

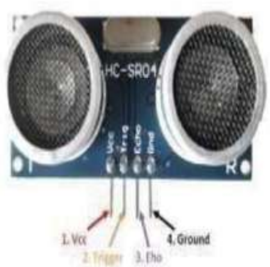

Figure 3.3: Ultra Sonic Sensor

Ultrasonic sensors measure distance by using ultrasonic waves. The sensor head emits an ultrasonic wave and receives the wave reflected back from the target. Ultrasonic Sensors measure the distance to the target by measuring the time between the emission and reception. An optical sensor has a transmitter and receiver, whereas an ultrasonic sensor uses a single ultrasonic element for both emission and reception.

3.4 Battery

A rechargeable battery, storage battery, or secondary cell (formally a type of energy accumulator), is a type of electrical battery which can be charged, discharged into a load, and recharged many times, as opposed to a disposable or primary battery, which is supplied fully charged and discarded after use.

Figure 3.4: Battery

3.5 Jumper Wires

Figure 3.5: Jumper wires

A jumper is a tiny metal connector that is used to close or open part of an electrical circuit. It may be used as an alternative to a dual in-line package (DIP) switch. A jumper has two or more connecting points, which regulate an electrical circuit board.

Chapter 4

SOFTWARE OF THE SYSTEM

Arduino is an open-source electronics platform based on easy-to- use hardware and software. Arduino boards are able to read inputs-light on a sensor, a finger on a button, or a Twitter message- and turn it into an output-activating a motor, turning on an LED, publishing something online. You can tell your board what to do by sending a set of instructions to the microcontroller on the board. To do so you use the Arduino Programming language (based on Wiring), and the Arduino software (IDE), based on Processing. Arduino's versatility and user-friendly nature have made it a popular choice for educators, students, hobbyists, and professionals alike. Its open-source framework encourages collaboration and innovation, allowing users to share their projects and ideas within a vibrant community. With a variety of boards and modules available, Arduino can be used for numerous applications, from simple DIY projects to complex industrial automation systems. Additionally, the extensive library support and vast online resources make it easy to find solutions and guidance for almost any project. Whether you're building a home automation system, a robotic arm, or a smart dustbin, Arduino provides a robust and flexible platform to bring your ideas to life.

4.1 Arduino IDE

Figure 4.1: Arduino IDE

The IDE (Integrated Development Environment) is a special program running on computer that allows one to write sketches for the Arduino board in a simple language modeled after the Processing language. To program the Arduino microcontroller Arduino coding language was used. The Arduino language is based on C/C++ and the most basic executable program only needs two functions a setup() and a loop(), to run. In the setup() function variables, pin modes, serial communication, etc are initialized. This function only runs once. The loop() function is where one write the actual code. As the name implies, the loop() function loops continuously until the device is powered off. Simple as it may sound; it is possible to write complex programs using the above described structure.

Chapter 5

Flow Chart

5.1 Explanation of Flowchart

The provided flowchart represents a simple control flow for a smart dustbin system using an Arduino

- Start:

  This is the initial state of the system where the program begins its execution.

- Check the Distance:

  This step involves using an ultrasonic sensor to measure the distance of an object from the dustbin. The sensor constantly checks for the presence of an object.

- If (distance $<$ 50) :

  This is a decision point where the system checks if the measured distance is less than 50 cm (or any predefined threshold distance). This threshold represents the proximity of an object (such as a hand or trash) to the dustbin.

- Open the Lid:

  If the distance is found to be less than the threshold (i.e., an object is detected within the range), the system will trigger a servo motor to open the lid of the dustbin.

- Stop:

  This indicates the end of this particular flow. In a real-world application, this might represent a pause before the system returns to the starting state to check the distance again, or it might represent a waiting period for the lid to close again.

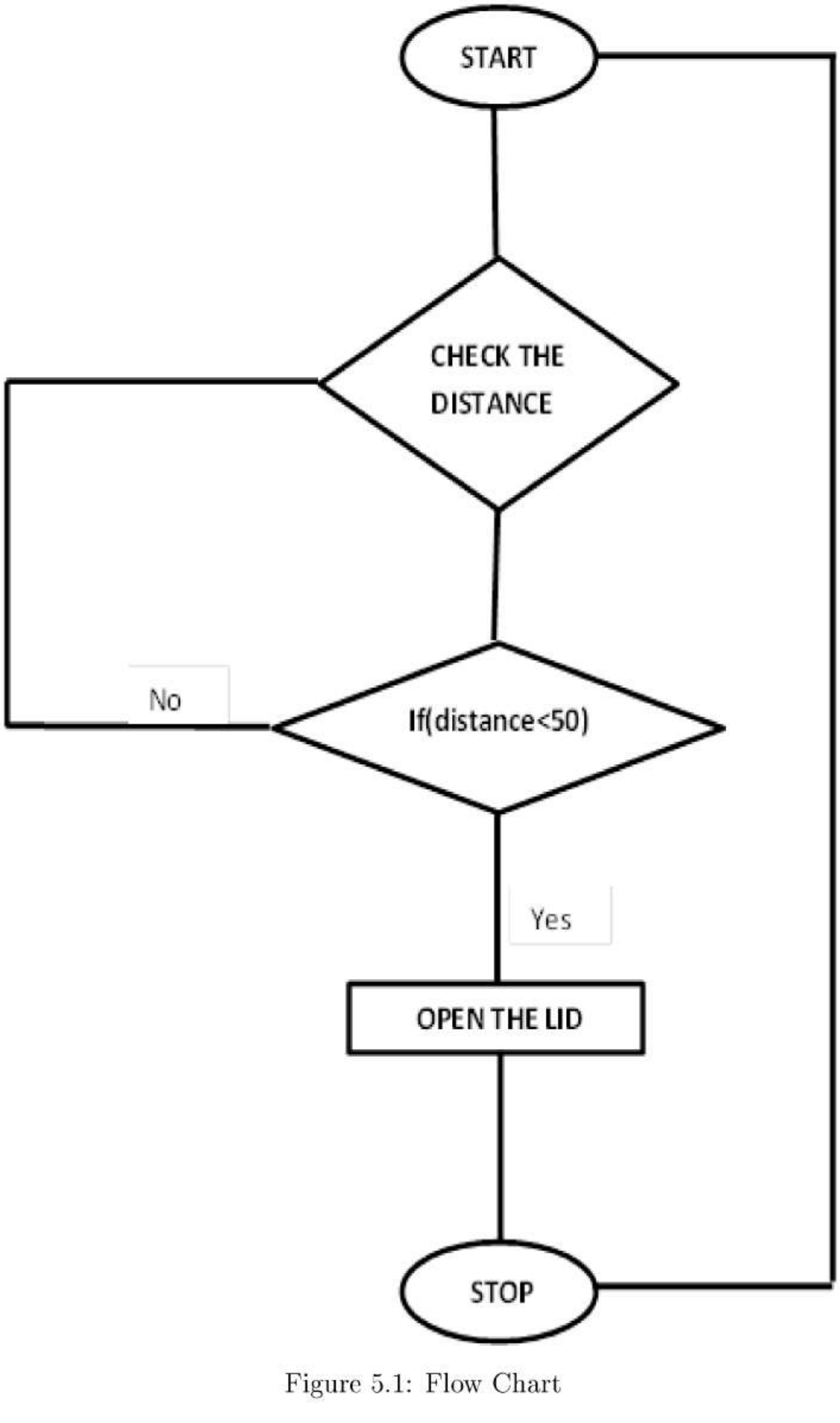

Figure 5.1: Flow Chart

Chapter 6

Referal Code

```
include
//servo library Servo servo;
int trigPin = 5;
int echoPin = 6;
int servoPin = 7;
int led= 10;
long duration, dist, average;
long aver[3]; //array for average
void setup()
Serial.begin(9600);
servo.attach(servoPin);
pinMode(trigPin, OUTPUT);
pinMode(echoPin, INPUT);
servo.write(0); //close cap on power on
delay(100);
servo.detach();
void measure()
digitalWrite(10,HIGH);
digitalWrite(trigPin, LOW);
delayMicroseconds(5);
digitalWrite(trigPin, HIGH);
delayMicroseconds(15);
digitalWrite(trigPin, LOW);
pinMode(echoPin, INPUT);
```

```
duration = pulseIn(echoPin, HIGH);
dist = (duration/2) / 29.1; //obtain distance
void loop()
for (int i=0;i<=2;i++) //average distance
measure();
aver[i]=dist;
delay(10); //delay between measurements
dist=(aver[0]+aver[1]+aver[2])/3;
if ( dist<50 )
//Change distance as per your need servo.attach(servoPin);
delay(1);
servo.write(0);
delay(3000);
servo.write(150);
delay(1000);
servo.detach();
Serial.print(dist);
```

Chapter 7

Working

7.1 Steps involved in working

1. Setup:

Connect the ultrasonic sensor, servo motor, and battery to the Arduino Uno using jumper wires.

2. Programming:

Write and upload code to the Arduino to control the sensor and motor.

3. Testing:

Ensure the sensor accurately detects objects and the motor responds correctly.

Final Assembly:

Integrate all parts into a functional smart dustbin.

Figure 7.1: Model of Dustbin

7.2 Working of the Model

1. After wiring and attaching all the devices and setting up to the Smart Dustbin, now observe all the important setup whether they are well connected or something missed.
2. After connection set up now next step is to submit/upload code in Arduino and supply power to the circuit.
3. When system is powered ON, Arduino keeps monitoring for any things that come near the sensor at give range.
4. When Ultrasonic sensor detect any object for example like hand or others, here Arduino calculates its distance and if it less than a certain predefined value than servo motor get activate first and with the support of the extended arm of the lid.
5. Lid will open for a given time than it will automatically close.[3]

Figure 7.2: Working of dustbin

Chapter 8

Result and Discussions

8.1 Advantages

1. Hygienic: Reduces the spread of germs with touch-free operation.

2. Convenient: Opens and closes automatically for easy waste disposal.

3. Odor Control: Keeps the lid closed, containing unpleasant smells.

4. User-Friendly: Simple and accessible for all users.

5. Efficient: Consumes minimal power and operates effectively.

6. Cost Savings: Reduced waste collection frequencies, lower fuel consumption, and extended bin lifetimes lead to significant cost savings.

7. Environmental Benefits: Encourages sustainable waste management practices, reducing waste sent to landfills and supporting a circular economy.

8. Improved Citizen Experience: Smart dustbins enhance the quality of life in urban areas, making them cleaner, healthier, and more livable.

9. Scalability: Can be easily scaled up for deployment in larger urban areas or scaled down for household use.

10. Reduced Labor: Automation reduces the need for manual intervention, lowering labor costs and minimizing human error.

11. Educational Tool: Serves as an excellent educational tool for teaching principles of electronics, programming, and environmental science.

12. Enhanced Safety: Reduces the need for frequent handling of waste, decreasing the risk of injuries and exposure to hazardous materials.

8.2 Application

1. Homes: Ideal for kitchens and bathrooms to maintain cleanliness.

2. Offices: Enhances hygiene in communal areas.

3. Public Places: Useful in malls, parks, and airports.

4. Healthcare: Keeps hospital and clinic environments sterile.

5. Restaurants: Ensures cleanliness in food preparation areas.

6. Urban Waste Management: Efficient waste collection, reduced waste overflow, and improved public health.

7. Smart Cities: Integration with city infrastructure for optimized waste management and data analysis.

8. Research and Development: Studying waste management patterns, behavior, and optimization strategies.

9. IoT Applications: Integration with other smart devices and sensors for a comprehensive smart city infrastructure.

10. Educational Institutions: Maintains hygiene in schools and universities, promoting a clean learning environment.

11. Government Buildings: Promotes a clean and professional environment in public offices and facilities.

12. Museums and Cultural Sites: Preserves cleanliness in areas with high foot traffic, protecting exhibits and artifacts.

Chapter 9

Conclusion

Here we are going to make an evolution changes toward cleanliness. The combination of intelligent waste monitoring and trash compaction technologies, smart dustbins are better and shoulders above traditional garbage dustbin. It is equipped with smart devices like sensor Arduino etc. Lid of the dustbin will automatically open when an object comes near to the dustbin and after certain time period it will close the lid. For social it will help toward health and hygiene, for business for we try to make it affordable to many as many possible. So that normal people to rich people can take benefit from it. Believe this will bring something changes in term of cleanliness as well technology. [4]

This smart dustbin represents a significant leap forward in our approach to waste management, combining cutting-edge technology with practical application to address hygiene and efficiency. By automating the waste disposal process, it minimizes human contact, thereby reducing the spread of germs and maintaining a cleaner environment. The affordability factor ensures that this innovative solution is accessible to a broad spectrum of users, from individual households to large-scale public spaces. Additionally, the integration of smart monitoring systems allows for real-time data collection, optimizing waste collection schedules and reducing operational costs. Overall, this project not only enhances cleanliness but also promotes a more sustainable and technologically advanced society.

9.1 Future Scope using Smart Dustbin

1. The future enhancement of this dustbin can be done by connecting this dustbin to the cloud and stores that who threw the garbage in the dustbin using RFID reader and RFID Tags for safety purpose.

2. The dustbin contains electrical wires, when liquid garbage is thrown into the dustbin it may lead to short circuit, the future work can be done by using liquid detecting sensors and avoids throwing liquid garbage into the dustbin.

3. Waste Collected in the Dustbin can also be segregated using the sensors and Integration with waste collection services to automate collection schedules based on bin's fill level.

4. Development of mobile apps to notify users about the nearest available bins and provide information on recycling practices.

5. Add sensors to notify when the bin is ful and Develop an app for remote monitoring and control 6. AI Integration: Incorporate artificial intelligence to analyze waste types and patterns, providing insights for improving waste management strategies

7. Public Awareness Campaigns: Develop educational campaigns and materials to promote the use of smart dustbins and sustainable waste management practices.

8. Waterproof Design: Ensure the dustbin is fully waterproof, allowing for safe usage in outdoor environments and during adverse weather conditions.

9. Emergency Alerts: Integrate emergency alert systems that can notify authorities in case of vandalism or suspicious activities around the dustbin.

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